

Introduction to Electrodynamics – David J. Griffiths

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1 Mathematical Review

1.1 Vector Analysis

A review of helpful vector stuff.

Scalar triple product: $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$.

- $\mathbf{B} \times \mathbf{C}$ is just the area spanned by $\mathbf{B}$ and $\mathbf{C}$, while $A \cos \theta$ is the altitude of $\mathbf{A}$ w.r.t. plane of $\mathbf{B}$ and $\mathbf{C}$
- Thus the scalar triple product is just the volume of the parallelepiped spanned by $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$.
- Consequently (order matters),

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

- You can swap dot and cross.

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$$

Vector triple product: Simplify using “BAC-CAB” rule.

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

With $\cdot$ and $\times$, there are 2 ways to construct a scalar (fg or $\mathbf{A} \cdot \mathbf{B}$) and 2 ways to construct a vector ($f\mathbf{A}$ or $\mathbf{A} \times \mathbf{B}$). This leads to 6 product rules.

1. $\nabla(fg) = f\nabla g + (\nabla f)g$
2. $\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A})$
 $+ (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
3. $\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
4. $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
5. $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$
6. $\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$
 $+ (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$

Quotient rules follow immediately from product rules.

Repeated application of the gradient, divergence, and curl result in further identities.

1. Divergence of gradient: the **Laplacian**

$$\nabla \cdot (\nabla T) \equiv \nabla^2 T = \frac{\partial^2}{\partial x^2} T_x + \frac{\partial^2}{\partial y^2} T_y + \frac{\partial^2}{\partial z^2} T_z$$

2. Curl of gradient: always 0.

$$\nabla \times (\nabla T) = 0$$

3. Gradient of divergence: $\nabla(\nabla \cdot \mathbf{v})$, not the Laplacian.

4. Divergence of curl: always 0.

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0$$

5. Curl of curl: use the following identity.

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}$$

1.2 Integral Calculus

Three important theorems:

1. Fundamental theorem of calculus for gradients:

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

2. The divergence theorem: Essentially “total flux $\rightarrow$ surface integral”

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}$$

3. Stokes’ theorem: Makes sense if you think “little curls within surface $\rightarrow$ integral over boundary”

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\partial S} \mathbf{v} \cdot d\mathbf{l}$$

1.3 Curvilinear Coordinates

For spherical coordinates, use θ as the polar angle (from z) and ϕ as the azimuthal angle (x - y plane).

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

- Infinitesimal line element: $d\mathbf{l} = (dr)\hat{r} + (r d\theta)\hat{\theta} + (r \sin \theta d\phi)\hat{\phi}$
- Infinitesimal volume element (Jacobian, anybody?): $dV = r^2 \sin \theta dr d\theta d\phi$

Gradient, divergence, and curl (which is a pain) are different for spherical coordinates. Consult Griffiths' book (p. 42 or Appendix A) because it's 4:30 am and I'm tired.

Cylindrical coordinates are easier. ϕ is the same as the azimuthal angle. z is the "height."

$$x = s \cos \phi, \quad y = s \sin \phi, \quad z = z$$

- Infinitesimal line element: $d\mathbf{l} = (ds)\hat{s} + (s d\phi)\hat{\phi} + (dz)\hat{z}$
- Infinitesimal volume element: $dV = s ds d\phi dz$

1.4 The Dirac Delta

If we take the divergence of $\mathbf{v} = \hat{r}/r^2$, it's 0. If we apply the divergence theorem,

$$\begin{aligned} \int_V (\nabla \cdot \mathbf{v}) d\tau &= \oint_S \mathbf{v} \cdot d\mathbf{a} \\ 0 &\stackrel{?}{=} \int \frac{1}{R^2} \hat{r} \cdot R^2 \sin \theta d\theta d\phi \hat{r} \\ 0 &\neq 4\pi \end{aligned}$$

The problem is that $\mathbf{v}$ does not have continuous partials. It blows up at $\mathbf{r} = 0$. To keep the divergence theorem consistent, we'll later see:

$$\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi \delta^3(\mathbf{r}) \longrightarrow \int (\nabla \cdot \mathbf{v}) d\tau = 4\pi$$

Dirac delta function in 1D: Technically a "distribution," not a function. Defined as

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(x) dx &= 1 \\ \delta(x) &= \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases} \end{aligned}$$

The most important property, used over and over again, will be:

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a)$$

Holy measure theory, batman.

- We can also define a **step function** $\theta(x)$ such that

$$\delta(x) = \frac{d\theta}{dx}, \quad \theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

- The Dirac delta in 3D is just a repeated application of the Dirac delta in 1D. But there's the interesting result of

$$\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi\delta^3(\mathbf{r})$$

Alternatively, $\nabla \cdot \left(\nabla \left(\frac{1}{r} \right) \right) = -4\pi\delta^3(\mathbf{r}) \rightarrow \nabla^2 \frac{1}{r} = -4\pi\delta^3(\mathbf{r})$

1.5 The Helmholtz Theorem

Maxwell's equations treat $\mathbf{E}$ and $\mathbf{B}$ in terms of their divergence and curl.

Loosely, the **Helmholtz theorem** says that a “smooth” vector field with boundary conditions (e.g. decaying quickly as $r \rightarrow \infty$) can be uniquely defined in terms of the divergence-free component and a curl-free component.

For example, since the divergence of a curl is 0 and the curl of a gradient is 0, we can always reduce a field $\mathbf{F}$ to

$$\mathbf{F} = -\nabla\phi + \nabla \times \mathbf{A}$$

You may recognize these as the scalar and vector potentials.

2 Electric Fields

2.1 Key Equations

Coulomb's law (for source charge q and test charge Q):

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}$$

$\epsilon_0 \approx 8.85 \times 10^{-12} C^2/(N \cdot m^2)$ is the permittivity of free space.

For multiple charges, we could simply say $\mathbf{F} = Q\mathbf{E}$ where the **electric field** $\mathbf{E}$ is defined by the charges

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i^2} \hat{r}_i$$

Or for a continuous charge distribution (line, surface, volume, etc.)

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq_i}{r_i^2} \hat{r}_i$$

Just make sure to get the infinitesimal volume element, surface element, etc. right.

2.2 Gauss's Law

The flux of $\mathbf{E}$ through a surface S is a measure of the electric **field lines** passing through S .

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{a}$$

Since field lines emanating from, e.g. a positive charge within S must either intersect S or terminate in some negative charge within S , the flux through S is a measure of the charge within. Hence **Gauss's law**:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \equiv \Phi_E$$

- Requires a closed surface
- Heavily simplifies $\mathbf{E}$ calculations. Common Gaussian surfaces to use are the sphere, cylinder, and pillbox.

The differential form of Gauss's law follows from the divergence theorem:

$$\begin{aligned} \oint \mathbf{E} \cdot d\mathbf{a} &= \int_V (\nabla \cdot \mathbf{E}) dV \\ Q_{\text{enc}} &= \int_V \rho dV \\ \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \end{aligned}$$

There's also a neat result for the curl of $\mathbf{E}$. E.g. for a point source,

$$\begin{aligned} \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}, \quad d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}} \\ \oint_{\partial S} \mathbf{E} \cdot d\mathbf{l} &= \int_S (\nabla \times \mathbf{E}) dS \\ \oint_{\partial S} \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr &= 0 = \int_S (\nabla \times \mathbf{E}) dS \\ \nabla \times \mathbf{E} &= 0 \end{aligned}$$

(For those of you who know Maxwell's equations – no, there's no conflict. This is the case of an electric field from static charges, not an electric field induced by a changing magnetic field.)

2.3 Electric Potential

Since $\nabla \times \mathbf{E} = 0$, by Stokes' we have $\oint \mathbf{E} \cdot d\mathbf{l} = 0$. We can see that the integral is independent of path. Thus we can define a "potential":

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}$$

where $\mathcal{O}$ is the reference point. Potentials are most useful in terms of their difference:

$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}$$

From the fundamental theorem of calculus for gradients,

$$- \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = V(\mathbf{b}) - V(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} (\nabla V) \cdot d\mathbf{l}$$

$$\mathbf{E} = -\nabla V$$

Example: The electric potential inside & outside a shell of charge with radius R . By Gauss's law,

$$\text{Inside: } \mathbf{E} = 0$$

$$\text{Outside: } \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$$

Now we can start to find the potentials. Outside, we have

$$V(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} = - \frac{1}{4\pi\epsilon_0} \int_{\infty}^{\mathbf{r}} \frac{Q}{r'^2} d\mathbf{r}' = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

Inside the shell, meanwhile,

$$V(\mathbf{r}) = - \int_{\infty}^R \mathbf{E}_{\text{out}} \cdot d\mathbf{l} - \int_R^{\mathbf{r}} \mathbf{E}_{\text{in}} \cdot d\mathbf{l}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{R} - \int_R^{\mathbf{r}} 0 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

Note that the potential is constant but not 0. Also note that the potential inside depends on the charges outside.

So far we have $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ and $\nabla \times \mathbf{E} = 0$. Since $\nabla \cdot \mathbf{E} = \nabla \cdot (-\nabla V) = -\nabla^2 V$, we recover **Poisson's equation**:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

In the special case that there's no charge ($\rho = 0$), we have **Laplace's equation**:

$$\nabla^2 V = 0$$

2.4 Work and Energy in Electrostatics

Work in electrostatics:

$$W = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} = -Q \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = Q [V(\mathbf{b}) - V(\mathbf{a})]$$

$$W = Q\Delta V$$

The sign depends on whether we're acting along or against the direction of $\mathbf{E}$. Based on W , we can also see that **electric potential energy** is the minimum energy required to assemble a certain configuration.

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{j>i}^N \frac{q_i q_j}{r_{ij}}$$

To get a formula for continuous charge distributions, note that

$$W = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{j \neq i}^N \frac{q_i q_j}{r_{ij}} = \frac{1}{2} \sum_{i=1}^N q_i V(\mathbf{r})$$

The continuous-distribution case is simple enough, using $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$.

$$W = \frac{1}{2} \int V dq = \frac{1}{2} \int \rho V d\tau = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V d\tau$$

If we integrate by parts (go through details on this...) we get

$$\begin{aligned} W &= \frac{\epsilon_0}{2} \left[- \int \mathbf{E} \cdot (\nabla V) d\tau + \oint V \mathbf{E} \cdot d\mathbf{a} \right] \\ &= \frac{\epsilon_0}{2} \left(\int_V E^2 d\tau + \oint_S V \mathbf{E} \cdot d\mathbf{a} \right) \end{aligned}$$

At large distances, $E \propto 1/r^2$ and $V \propto 1/r$ while the surface grows as r^2 . By sending the whole volume integral to ∞ , we enclose all relevant charges while sending the surface integral to 0. Thus

$$W = \frac{\epsilon_0}{2} \int E^2 dV$$

The energy is stored in the $\mathbf{E}$ field with density $\frac{\epsilon_0}{2} E^2$ energy per unit volume.

2.5 Capacitors

In **insulators**, electrons can't wander too far from the atom. In **conductors**, electrons are free to move from atom to atom.

1. $\mathbf{E} = 0$ inside conductor: charges rearrange in least-energy configuration.
2. $\rho = 0$ inside conductor.
3. Net charges reside on surface, which is equipotential.
4. $\mathbf{E}$ just outside is perpendicular to surface.

We define the capacitance as $C = Q/V$, with V being the potential between two conductors.

1. E.g. for two parallel plates of area A and charges $-Q, +Q$, we have

$$\begin{aligned} \sigma &= \frac{Q}{A}, \quad \mathbf{E} = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}, \quad V = \frac{Q}{A\epsilon_0} d \\ C &= \frac{\epsilon_0 A}{d} \end{aligned}$$

2. E.g. for two concentric spherical shells with radii a , b ; charge $+Q$ on inner and $-Q$ on the outer, we have

$$\begin{aligned}\text{Electric field between shells: } \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \frac{Q}{\mathbf{r}^2} \hat{\mathbf{r}} \\ V &= - \int_a^b \frac{1}{4\pi\epsilon_0} \frac{Q}{\mathbf{r}^2} \hat{\mathbf{r}} \cdot d\mathbf{l} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) \\ C &= 4\pi\epsilon_0 \frac{ab}{(b-a)}\end{aligned}$$

“Charging up” a capacitor means moving electrons from the positive plate to the negative plate. The total work necessary (and thus maximum potential of capacitor) is

$$W = \frac{Q^2}{2C} = \frac{1}{2} CV^2$$

This is derived from $dW = dqV$, so $\frac{dW}{dq} = q/C$ and integrate.